

Preface

These are notes for a course on the representation theory of the symmetric group. This topic combines combinatorics, abstract algebra, and linear algebra to understand how representations of the symmetric group break into irreducible components. Second references include...

Contents

Contents	i
1 Representation of Finite Groups	1
1.1 Matrix Representations	1
Exercises	5

Chapter 1

Representation of Finite Groups

§ 1.1. Matrix Representations

Definition 1.1.1. Let G be a finite group. Let $\text{GL}_n(\mathbb{C})$ be the set of $n \times n$ invertible matrices over $\mathbb{C}$. A degree n (matrix) representation of G is a homomorphism $X : G \rightarrow \text{GL}_n(\mathbb{C})$.

Example 1.1.1. Let G be any group. Let $X : G \rightarrow \text{GL}_1(\mathbb{C})$ be defined by $X(g) = 1$ for all $g \in G$. This is the *trivial representation*.

Example 1.1.2. Let $X : \mathbb{Z}_2 \times \mathbb{Z}_2 \rightarrow \text{GL}_2(\mathbb{C})$ be defined by:

$$\begin{aligned} X((0,0)) &:= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ X((1,0)) &:= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ X((0,1)) &:= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

Note that $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ are their own inverses, as are $(1,0), (0,1) \in \mathbb{Z}_2 \times \mathbb{Z}_2$. The last element is predetermined as:

$$X((1,1)) := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Recall 1.1.1. The *sign* of a permutation is a function $\text{sign} : S_n \rightarrow \{\pm 1\}$ defined by $\text{sign}(\sigma) := (-1)^m$, where m is the number of transpositions in the permutations decomposition. For example, $(2\ 3\ 6\ 4) = (2\ 3)(3\ 6)(6\ 4)$, hence $\text{sign}((2\ 3\ 6\ 4)) = (-1)^3 = -1$.

Example 1.1.3. Let $X : S_n \rightarrow \text{GL}_1(\mathbb{C})$ be defined by $X(\sigma) := \text{sign}(\sigma)$. This is a representation, as the sign function is multiplicative. Indeed, if σ is a k -transposition and τ is a t -transposition, then $\sigma\tau$ is a $k+t$ transposition. This proves that $\text{sign}(\sigma)\text{sign}(\tau) = \text{sign}(\sigma\tau)$, as even + even = even, even + odd = odd, odd + even = odd, and odd + odd = even.

Recall 1.1.2. A *group action* of a group G on a set A is a map $G \times A \rightarrow A$ (written $g.a$ for all $g \in G$ and $a \in A$) satisfying the following properties:

- $g_1.(g_2.a) = (g_1g_2).a$ for all $g_1, g_2 \in G$, $a \in A$, and

- $1.a = a$ for all $a \in A$.

Example 1.1.4. Let $S = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$ and $G = S_4$. A group action of S_4 on S can be defined by the group element of S_4 permuting a two-element set of S . For example:

$$\begin{aligned}(1\ 2).\{1, 3\} &:= \{2, 3\}, \\ (1\ 2\ 3\ 4).\{1, 2\} &:= \{2, 3\}.\end{aligned}$$

This is indeed a group action, as:

$$\begin{aligned}(1\ 2).((1\ 2\ 3\ 4).\{1, 2\}) &= (1\ 2).\{2, 3\} = \{1, 3\} \\ ((1\ 2)(1\ 2\ 3\ 4)).\{1, 2\} &= (2\ 3\ 4).\{1, 2\} = \{1, 3\}.\end{aligned}$$

Example 1.1.5. Given S from the previous example, we can define an action on S by $\mathbb{Z}_4$. First, note that if σ is a 4-cycle, then $\mathbb{Z}_4 \rightarrow \langle \sigma \rangle$ is an isomorphism given by $[k] \mapsto \sigma^k$. This means if we fix a 4-cycle, we can define an action on S by $\mathbb{Z}_4$. For example, we can define:

$$\begin{aligned}[0].\{1, 2\} &:= (1).\{1, 2\} = \{1, 2\} \\ [1].\{1, 2\} &:= (1\ 2\ 3\ 4).\{1, 2\} = \{2, 3\} \\ [2].\{1, 2\} &:= (1\ 2\ 3\ 4)^2.\{1, 2\} = (1\ 2\ 3\ 4).\{2, 3\} = \{3, 4\} \\ [3].\{1, 2\} &:= (1\ 2\ 3\ 4)^3.\{1, 2\} = \{1, 4\}.\end{aligned}$$

Example 1.1.6. Let:

$$V := \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 + x_2 + x_3 = 0 \right\}$$

be a subspace of $\mathbb{C}^3$. Given the set information, we can write $x_1 = -x_2 - x_3$, hence there are two-free variables. If $x_2 = 1$ and $x_3 = 0$, then one of our basis elements is $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. If $x_2 = 0$

and $x_3 = 1$, then our other basis element is $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$.

The group S_3 acts on V by permuting coordinates. For example:

$$(1\ 2\ 3).\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix}.$$

Since V is a two-dimensional subspace of $\mathbb{C}$ with basis $\left\{ v_1 := \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, v_2 := \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$, we can construct a representation $X : V \rightarrow \text{GL}_2(\mathbb{C})$ as follows. First, note that:

$$\begin{aligned}(1).v_1 &= v_1 = (1)v_1 + (0)v_2, \\ (1).v_2 &= v_2 = (0)v_1 + (1)v_2.\end{aligned}$$

Hence define:

$$X((1)) := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Similarly,

$$\begin{aligned} (1\ 2).v_1 &= (1\ 2). \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = -v_1 = (-1)v_1 + (0)v_2, \\ (1\ 2).v_2 &= (1\ 2). \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = -v_1 + v_2 = (-1)v_1 + (1)v_2. \end{aligned}$$

Thus:

$$X((1\ 2)) := \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix}.$$

This can be done for all 9 elements of S_3 . One must then verify by 36 different computations to show that X is indeed a group homomorphism. This is tedious, and instead we will show that a group action *induces* a group homomorphism.

Theorem 1.1.1. *Let G act on $S := \{s_1, \dots, s_n\}$. Define $X : G \rightarrow \text{GL}_n(\mathbb{C})$ by:*

$$X(g)_{ij} := \begin{cases} 1, & g.s_j = s_i \\ 0, & \text{otherwise} \end{cases}.$$

Then X is a degree n representation.

Proof. Let V be the (free) vector space over $\mathbb{C}$ with basis S . Note that linear maps are defined by their action on a basis. Fix an element $g \in G$. We define a linear map that sends s_j to $g.s_j$ for all $j = 1, \dots, n$. Then the matrix for this linear map is X . Since $e.s_j = s_j$ and $g.(h.s_j) = (gh).s$, then $X(e) = I_n$ and $X(gh) = X(g)X(h)$. Thus X is a homomorphism. ■

Example 1.1.7. Consider the set S from Example ?? . Let $s_1 := \{1, 2\}$, $s_2 := \{1, 3\}$, and so on. Given the representation defined in Theorem ??, if we want to compute $X((4\ 3\ 2\ 1))$ then consider:

$$\begin{aligned} (4\ 3\ 2\ 1).s_1 &= (4\ 3\ 2\ 1).\{1, 2\} = \{1, 4\} = s_3, \\ (4\ 3\ 2\ 1).s_2 &= (4\ 3\ 2\ 1).\{1, 3\} = \{2, 4\} = s_5, \\ (4\ 3\ 2\ 1).s_3 &= (4\ 3\ 2\ 1).\{1, 4\} = \{3, 4\} = s_6, \\ &\vdots \end{aligned}$$

Since $(4321).s_1 = s_3$, there will be a 1 in the third row and first column of $X((4321))_{ij}$.
Hence:

$$X((4321)) = \begin{matrix} & s_1 & s_2 & s_3 & s_4 & s_5 & s_6 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \\ s_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}.$$

Exercises

Exercise 1.

- (a) Let E be an $n \times n$ matrix over $\mathbf{C}$ such that $E \neq I_n$, $E \neq 0$, and $E^2 = E$. Show that there is an $n \times n$ matrix T such that:

$$E = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T$$

for some k .

- (b) Let G be a group and X be a function mapping G into the set of (possible singular) $n \times n$ matrices over $\mathbf{C}$ such that $X(e) \neq 0$ and $X(gh) = X(g)X(h)$ for all $g, h \in G$. Show that there is an $n \times n$ matrix T and a matrix representation Y of G such that:

$$X(g) = T^{-1} \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} T$$

for all $g \in G$.

Proof. (a) Since E is a projection, its eigenvalues are $\lambda = 0$ and $\lambda = 1$. Let $E_0 := \ker(E)$ and $E_1 := \ker(E - I_n)$ denote the eigenspaces of $\lambda = 0$ and $\lambda = 1$. Note that $Ev \in E_1$, as:

$$\begin{aligned} (E - I_n)(Ev) &= E^2v - Ev \\ &= 0. \end{aligned}$$

Similarly, $v - Ev \in E_0$ since:

$$\begin{aligned} E(v - Ev) &= Ev - E^2v \\ &= 0. \end{aligned}$$

Since every $v \in V$ can be written as $(v - Ev) + Ev$, this means $V = E_0 + E_1$. If $x \in E_0 \cap E_1$, then $Ex = 0$ and $Ex = x$. It must be the case that $x = 0$, hence $V = E_0 \oplus E_1$. Since V is the direct sum of eigenspaces, E is diagonalizable. Assuming that $\dim(E_1) = k$ and $\dim(E_0) = n - k$,

there exists a matrix T such that $E = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T$.

- (b) Note that $X(e)^2 = X(e)X(e) = X(e)$. Hence by part (a) there exists an $n \times n$ matrix T such that:

$$X(e) = T^{-1} \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} T.$$

Note that:

$$T X(g) T^{-1} = (T X(g) T^{-1})(T X(e) T^{-1}) = (T X(g) T^{-1}) \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}.$$

Similarly:

$$T X(g) T^{-1} = (T X(e) T^{-1})(T X(g) T^{-1}) = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} (T X(g) T^{-1}).$$

Since $(T X(g) T^{-1}) \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} (T X(g) T^{-1})$, it must be the case that $T X(g) T^{-1} = \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix}$ for some $Y(g) \in \text{Mat}_k(\mathbf{C})$. It remains to show that Y is a representation. Note that:

$$\begin{bmatrix} Y(\epsilon) & 0 \\ 0 & 0 \end{bmatrix} = T X(\epsilon) T^{-1} = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}.$$

Thus $Y(\epsilon) = I_k$. Similarly:

$$\begin{aligned} \begin{bmatrix} Y(gh) & 0 \\ 0 & 0 \end{bmatrix} &= T X(gh) T^{-1} \\ &= (T X(g) T^{-1})(T X(h) T^{-1}) \\ &= \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} Y(h) & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} Y(g)Y(h) & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Thus $Y(gh) = Y(g)Y(h)$. It remains to show that $\text{im } Y = GL_n(\mathbf{C})$. Indeed, for any $Y(g) \in \text{im } Y$ we have $Y(\epsilon) = Y(gg^{-1}) = Y(g)Y(g^{-1})$, hence $Y(g)$ does have an inverse. Thus, we've shown that there exists an $n \times n$ matrix T and matrix representation Y of G such that:

$$X(g) = T^{-1} \begin{bmatrix} Y(g) & 0 \\ 0 & 0 \end{bmatrix} T \quad \blacksquare$$

Exercise 2. For $\sigma = \sigma_1 \dots \sigma_n \in S_n$, written in one line notation, $\text{inv } \sigma$ is the number of pairs $i < j$ such that $\sigma_i > \sigma_j$. Define $\text{sign } \sigma := (-1)^{\text{inv } \sigma}$. Show that switching the position of any two integers in σ changes $\text{inv } \sigma$ by an odd number. This fact implies:

$$\text{sign}(\sigma\tau) = \text{sign } \sigma \text{sign } \tau,$$

where τ is any transposition. Since any permutation can be written as a product of transpositions, this means that the sign representation $X(\sigma) = \text{sign}(\sigma)$ is indeed a matrix representation of S_n .

Proof. Let τ be obtained from σ by swapping the entries in positions $p < q$. After the swap, suppose $\sigma_p < \sigma_q$. Let $m = \text{card}\{r : p < r < q \text{ and } \sigma_p < \sigma_r < \sigma_q\}$. Any inversions involving neither p nor q is unchanged, hence the total number of inversions is unchanged. Entries outside the interval from p to q also leave the total number of inversions unchanged. The pair (p, q) was not an inversion in σ , but becomes one in τ , contributing +1. Now consider $p < r < q$. If $\sigma_p < \sigma_r < \sigma_q$, then after the swap (p, r) and (r, q) are inversions. Thus each r contributes +2. If $\sigma_r < \sigma_p$ or $\sigma_r > \sigma_q$, then in either case the pairs contribute one inversion both before and after the swap, so there is no change. Thus:

$$\text{inv}(\tau) - \text{inv}(\sigma) = 1 + 2m,$$

which is odd. If $\sigma_p > \sigma_q$, the same argument gives $-(1 + 2m)$ which is also odd.

If $\sigma \in S_n$ and τ is a transposition, then:

$$\text{sign}(\sigma\tau) = \quad \blacksquare$$